Dimitri Lerévérend

Ph.D. student in Privacy-Preserving Decentralized Learning

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Profile

I am a Ph.D. student in privacy-preserving decentralized learning at INRIA Rennes. My research focuses on differential privacy (DP) for decentralized learning, with a particular emphasis on noise correlation. I work on both theory and practice to design approaches that are scalable, tractable, and supported by strong DP guarantees.

Education

2023–present **Ph.D. in Computer Science**, *INRIA Rennes*, Rennes, Advisors: Davide Frey, Romaric Gaudel, François Taïani
Exploring differential privacy (DP) in decentralized learning via noise correlation, from theoretical foundations to practical methods.

2022–2023 **Master's degree (M2 SIF) in Computer Science**, *ENS Rennes*, Rennes, France
Focus on machine learning, image representation, and complex systems.

2021–2022 **Preparation for the "Agrégation d'informatique"**, *ENS Rennes*, Rennes, France
Focus on teaching and computer science fundamentals: algorithms, architecture, operating systems, networking, and databases.

2019–2020 **Bachelor's degree in Computer Science**, *École Normale Supérieure de Rennes*, Rennes, France

Work experience

2023–present **Ph.D. Student**, *INRIA Rennes*, Davide Frey & Romaric Gaudel & François Taïani.
Conducting research on differential privacy for decentralized learning and collaborating with researchers across multiple institutions.

2025 May—July **Research Intern**, *Decentralized learning convergence under fragmentation*, Sacs Lab, EPFL, Anne-Marie Kermarrec
Studied theoretical convergence rates of decentralized learning with model fragmentation.

2023 January — July **Research Intern**, *Privacy protection through private averaging*, Team WIDE, INRIA Rennes, Davide Frey, François Taïani, Romaric Gaudel
Studied privacy properties of decentralized machine learning and the effects of private averaging using PyTorch.

2021 May—July **Research Intern**, *Adapting Public Announcement Logic update mechanisms to Dynamic Epistemic Logic*, D7, IRISA Rennes, Guillaume Aucher

2020–2021 **Research project**, *Feature extraction for high dynamic range images*, Percept, INRIA Rennes, Olivier Le Meur, Mathieu Chambe
Built a neural network for HDR image feature extraction in TensorFlow.

2020 May—July **Research Intern**, *Visualization of echo state networks*, Mnemosyne, INRIA Bordeaux, Nicolas Rougier
Developed a spatialized ESN and visualized activity using Voronoi diagrams.

Publications

[1] Aurélien Bellet, Edwige Cyffers, Davide Frey, Romaric Gaudel, **Dimitri Lerévérend***, and François Taïani. Unified privacy guarantees for decentralized learning via matrix factorization. *International Conference on Learning Representations*, 2026. URL https://iclr.cc/virtual/2026/poster/10010197. Code: github.com/dimiarbre/MFDL.

[2] Sayan Biswas, Davide Frey, Romaric Gaudel, Anne-Marie Kermarrec, **Dimitri Lerévérend***, Rafael Pires, Rishi Sharma, and François Taïani. Low-Cost Privacy-Preserving Decentralized Learning. *Proceedings on Privacy Enhancing Technologies*, 2025(3):451–474, 2025. ISSN 2299-0984. doi: 10.56553/popets-2025-0108. URL https://petsymposium.org/popets/2025/popets-2025-0108.php. Code: github.com/dimiarbre/ZIP-DL.

[3] Sayan Biswas, Antoine Boutet, Davide Frey, Romaric Gaudel, Rachid Guerraoui, Maxime Jacovella, Anne-Marie Kermarrec, **Dimitri Lerévérend**, François Taïani, and Martijn de Vos. Your neighbors know: Leveraging local neighborhoods for backdoor detection in decentralized learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. URL https://arxiv.org/abs/2605.19969.

[4] Sayan Biswas, Davide Frey, Romaric Gaudel, Nirupam Gupta, Anne-Marie Kermarrec, **Dimitri Lerévérend**, Rafael Pires, Rishi Sharma, François Taïani, and Martijn de Vos. Mosaic Learning: A Framework for Decentralized Learning with Model Fragmentation. (arXiv:2602.04352), February 2026. doi: 10.48550/arXiv.2602.04352. URL https://arxiv.org/abs/2602.04352.

* Primary contributor.

Diplomas/Certifications

2023 **Master's degree**, *University of Rennes*

2023 **Agrégation d'informatique**, *French national education*, Ranked 2nd.

Teaching

2024-2026 **PRI**, *M1*, Cyberschool
Course presenting privacy formally.

2024-2026 **Agrégation preparation**, *M1*, ENS Rennes
Course preparing students for the "Agrégation d'informatique", the national exam for teaching computer science in France.

2025-2026 **ISE**, *L1*, Cyberschool
Basic introduction to security with a mini-project.

2023-2025 **ACE**, *L1*, ISTIC
Course explaining algorithm complexity practically.

2024-2025 **Trustworthy AI**, *M1*, Master IA
Course introducing trustworthy AI: privacy and explainability.

2023-2024 **BDD**, *L2*, ISTIC
Course presenting databases.

Presentations

2026 **ICLR**, *Poster Presentation*, Rio

2026 **Transnumériques**, *Oral Presentation*, Rennes

2026 **YSP**, *Oral Presentation*, Paris

2025 **EurIPS**, *Workshop Presentation*, Copenhagen

Programming skills

Languages	Python, OCaml, C, C++, Java
Libraries	scikit-learn, TensorFlow, PyTorch, JAX, pandas, NumPy